The Journal of Open Source Software

GrIML: A Python package for investigating Greenland's ice-marginal lakes under a changing climate

Penelope R. How [1¶]

1 Department of Glaciology and Climate, Geological Survey of Denmark and Greenland (GEUS), Nuuk, Greenland ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Adam R. Jensen

Reviewers:

- @rwegener2
- @facusapienza21

Submitted: 22 January 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

The `GrIML` Python package is for processing classified water bodies from satellite imagery, and compiling classifications into a standardised inventory. It has been used to produce the Greenland ice-marginal lake inventory series, which maps the presence and extent of water bodies across Greenland that share a margin with the Greenland Ice Sheet and/or the surrounding ice caps and periphery glaciers since 2016 (How et al., 2025).

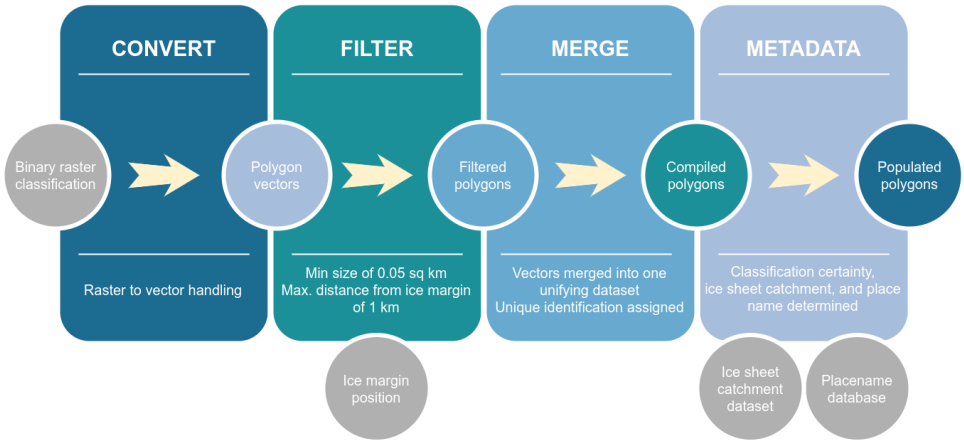

Figure 1: An overview of the GrIML Python package workflow for generating ice-marginal lake inventories. The boxes refer to the four main modules and processing stages (reading from left to right), the coloured circles represent intermediary outputs to each processing stage, and the grey circles represent additional inputs.

Initial rasterised binary classifications denoting water bodies can be inputted to `GrIML` to convert, filter, and merge into a unified ice-marginal lake vector dataset, populated with useful metadata and analysed with relevant statistical information (Figure 1).

Statement of need

`GrIML` meets four main needs of users in the remote sensing and cryospheric science communities:

1. Provide a usable workflow for processing rasterised water body classifications
2. Document the criteria for classifying an ice-marginal lake

3. Showcase a transparent workflow that, in turn, produces an open and reproducible ice-marginal lake dataset that adheres to the FAIR principles (Wilkinson et al., 2016)
4. Produce inventories that map the areal extent and abundance of ice-marginal lakes across Greenland, which demonstrate ice-marginal lake evolution under a changing climate

There have been many different approaches to classifying ice-marginal lakes with remote sensing techniques (Andreassen et al., 2022; Chen et al., 2021; Dømgaard et al., 2024; How et al., 2021; Rick et al., 2022; Wieczorek et al., 2023). Packages exist for handling satellite and spatial data, such as GrIML's two key dependencies, Geopandas (Van den Bossche & others, 2024) and Rasterio (Gillies & others, 2013--), as well as others such as SentinelHub (Sentinel Hub, 2024) and Google Earth Engine (Gorelick et al., 2017). Remote sensing object classification and post-processing routines are usually available in connection with scientific publications, however, few are available as open, deployable packages. The `GrIML` Python package addresses this gap, for the benefit of the future generation of ice-marginal lake inventories and for others in the scientific community to adapt and use themselves.

The 2016-2023 Greenland ice-marginal lake inventory series was produced with the `GrIML` Python package to address knowledge gaps regarding sea level contribution, and provide an overview of regions where lacustrine ice loss and processes are prominent (How et al., 2025, In Review). The dataset consists of a series of annual inventories, mapping the presence and extent of ice-contact lakes across Greenland (Figure 2). The annual inventory series spans the entirety of Greenland, including all terrestrial regions. Thus far, there are 8 annual inventories, covering 2016 to 2023, with one inventory for each year. The `GrIML` package will continue to be used to produce future additions to the Greenland ice-marginal lake inventory series; in which case a standardised and version-controlled processing pipeline is essential (How et al., 2025).

DRAFT

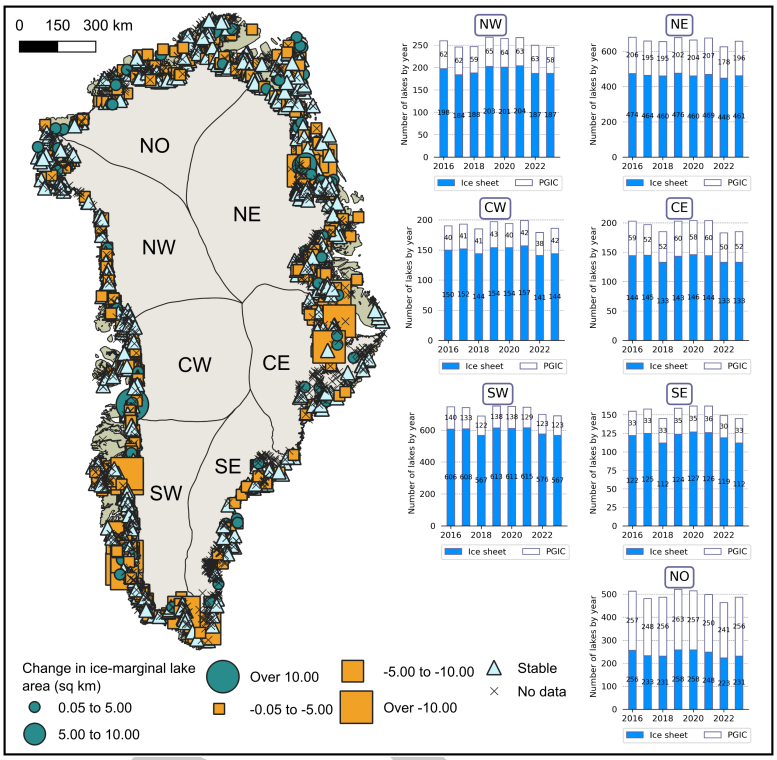

Figure 2: An overview of the Greenland ice-marginal lake inventory series, which maps the number of lakes and their size between the years of 2016 and 2023 (How et al., 2025). The map shows the location of each lake, with the symbol signifying area change over time, where green circles represent an increase in lake area, yellow squares represent a decrease in lake area, and white triangles represent an unchanged lake area. Crosshair points denote lakes where no area information is provided. The bar graphs show the change in number of lakes in each region. Base maps are provided by Moon et al. (2023) and region outlines are as defined by Mouginot & Rignot (2019).

Overview of package contents

`GrIML` has 4 main modules for processing from raster classifications to a vectorised inventory:

1. `Convert`: for converting binary raster classifications of water bodies to vectors
2. `Filter`: for filtering vectors by size and proximity to an object (i.e. an ice margin profile)
3. `Merge`: for merging vector classifications into one unified object (for example, when classifications cover a large area that needs to be split over multiple files)
4. `Metadata`: for populating vector classifications with various pieces of information, including identification number, placename, region, classification source, and certainty.

These 4 modules represent the 4 processing steps that should be executed in order to produce a standardised ice-marginal lake inventory. Each module contains the functions associated with each step (see Figure 3).

DRAFT

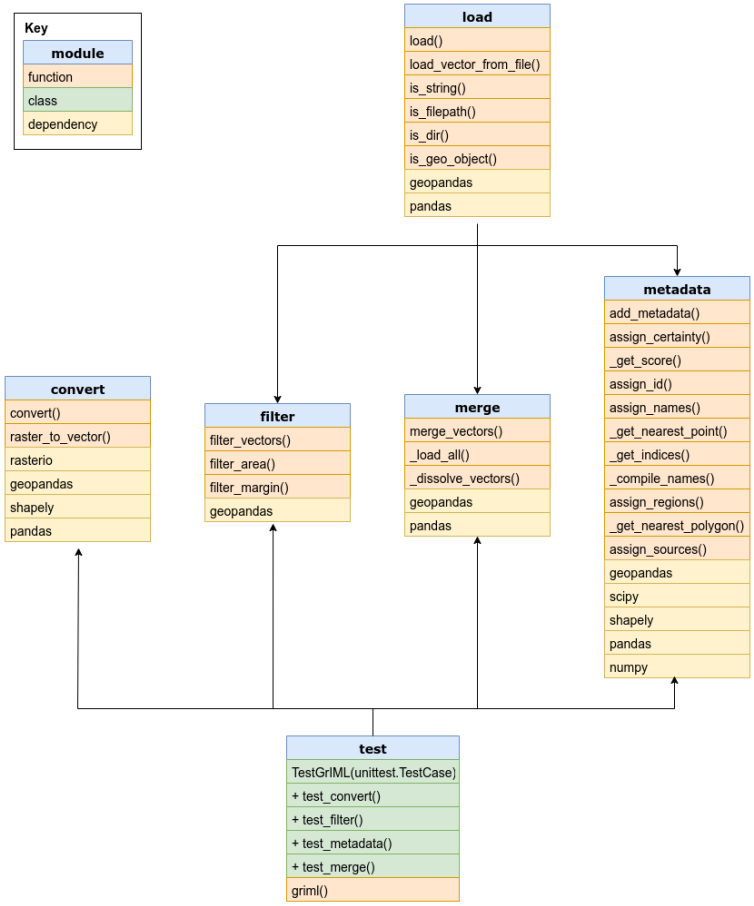

Figure 3: The GrIML Python package structure, including modules (blue), key module functions (orange), classes (green) and dependencies (yellow). Arrows between modules represent module dependencies, where the direction of the arrow indicates the dependency source (at the end of the arrow) and target (at the head of the arrow).

DRAFT

Usage

Installation

The `GrIML` Python package is compatible with Python 3.10, 3.11, and 3.12, and can be installed using pip:

```
$ pip install griml
```

Full installation instructions are provided at https://griml.readthedocs.io/en/latest/installation.html.

Tutorials and guides

Two sets of tutorials are provided with `GrIML` for 1) using the processing pipeline (https://griml.readthedocs.io/en/latest/tutorials-processing.html); and 2) handling the produced dataset (https://griml.readthedocs.io/en/latest/tutorials-data.html). The processing tutorials guide the user through each processing step from the initial raster classifications to a fully compiled ice-marginal lake dataset. The dataset tutorials provide an overview for how to handle and analyse an ice-marginal lake inventory produced with `GrIML`. This includes examples using the ice-marginal lakes dataset on the GEUS Dataverse (How et al., 2025), primarily handled

with Geopandas (Van den Bossche & others, 2024) to create spatial plots and time-series (see Figure 4).

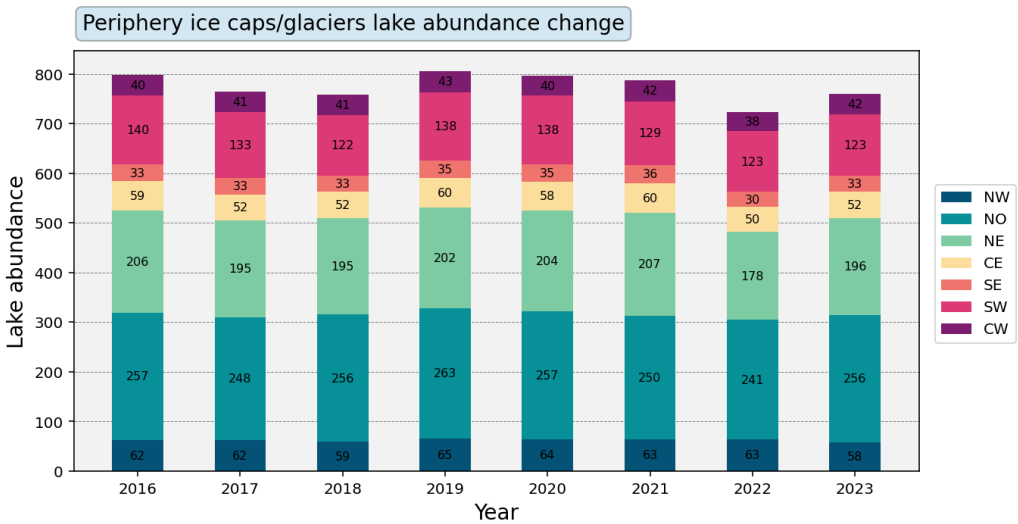

Figure 4: An example plot output from the dataset tutorial, where the change in the number of lakes across the ice-marginal lake inventory series is visualised. Each bar represents the number of lakes in one inventory year, with each coloured stage indicating the number of lakes for a given region.

Usage terms and conditions

If the `GrIML` package or the Greenland ice marginal lake inventory series are used to support results of any kind then we ask the authors to include references to the applicable publications:

- How, P. et al. (2025) *"Greenland Ice Marginal Lake Inventory annual time-series Edition 1"*. **GEUS Dataverse.** https://doi.org/10.22008/FK2/MBKW9N
- How, P. et al. (Submitted) *"The Greenland Ice-Marginal Lake Inventory Series from 2016 to 2023"*. **Earth System Science Data.**
- How, P. et al. (2021) *"Greenland-wide inventory of ice marginal lakes using a multi-method approach"*. **Scientific Reports** 11, 4481. https://doi.org/10.1038/s41598-021-83509-1

And include the following statement in the acknowledgements:

"Ice marginal lake data provided by the European Space Agency (ESA), and the Programme for Monitoring of the Greenland Ice Sheet (PROMICE) at the Geological Survey of Denmark and Greenland (GEUS) https://doi.org/10.22008/FK2/MBKW9N."

Acknowledgements

This work is funded by the ESA Living Planet Fellowship (4000136382/21/I-DT-lr) entitled "Examining Greenland's Ice Marginal Lakes under a Changing Climate". Further support was provided by PROMICE (Programme for Monitoring of the Greenland Ice Sheet) and GC-Net (Greenland Climate Network), which is funded by the Geological Survey of Denmark and Greenland (GEUS) within the Danish Ministry of Climate, Energy and Utilities. The Danish Finance Law currently supports GEUS for the continuation of PROMICE and GC-Net field and data activities. PROMICE and GC-Net are conducted in collaboration with DTU Space (Technical University of Denmark) and ASIAQ Greenland Survey.

DRAFT

References

Andreassen, L. M., Nagy, T., Kjøllmoen, B., & Leigh, J. R. (2022). An inventory of Norway's glaciers and ice-marginal lakes from 2018–19 Sentinel-2 data. *Journal of Glaciology*, *68*(272), 1085–1106. https://doi.org/10.1017/jog.2022.20

Chen, F., Zhang, M., Guo, H., Allen, S., Kargel, J. S., Haritashya, U. K., & Watson, C. S. (2021). Annual 30 m dataset for glacial lakes in High Mountain Asia from 2008 to 2017. *Earth System Science Data*, *13*, 741–766. https://doi.org/10.5194/essd-13-741-2021

Dømgaard, M., Kjeldsen, K., How, P., & Bjørk, A. (2024). Altimetry-based ice-marginal lake water level changes in Greenland. *Communications Earth and Environment*, *5*, 365. https://doi.org/10.1038/s43247-024-01522-4

Gillies, S., & others. (2013--). Rasterio: Geospatial raster i/o for Python programmers. *Mapbox*. https://github.com/rasterio/rasterio

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., & Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment*, *202*, 18–27. https://doi.org/10.1016/j.rse.2017.06.031

How, P., Messerli, A., Mätzler, E., Santoro, M., Wiesmann, A., Caduff, R., Langley, K., Bojesen, M. H., Paul, F., Kääb, A., & Carrivick, J. L. (2021). Greenland-wide inventory of ice marginal lakes using a multi-method approach. *Scientific Reports*, *11*, 4481. https://doi.org/10.1038/s41598-021-83509-1

How, P., Petersen, D., Karlsson, N. B., Kjeldsen, K. K., Raundrup, K., Messerli, A., Rutishauser, A., Carrivick, J. L., Lea, J. M., Fausto, R. S., Ahlstrøm, A. P., & Andersen, S. B. (2025). Greenland ice marginal lake inventory annual time-series Edition 1 [dataset]. *GEUS Dataverse*. https://doi.org/10.22008/FK2/MBKW9N

How, P., Petersen, D., Kjeldsen, K. K., Raundrup, K., Karlsson, N. B., Messerli, A., Rutishauser, A., Carrivick, J. L., Lea, J. M., Fausto, R. S., Ahlstrøm, A. P., & Andersen, S. B. (In Review). The Greenland ice marginal lake inventory series from 2016 to 2023. *Earth System Science Data*. https://doi.org/10.5194/essd-2025-18

Moon, T. A., Fisher, M., Stafford, T., & Thurber, A. (2023). QGreenland (v3). *National Snow and Ice Data Center*. https://doi.org/10.5281/zenodo.12823307

Mouginot, J., & Rignot, E. (2019). Glacier catchments/basins for the Greenland Ice Sheet. *Dryad*. https://doi.org/10.7280/D1WT11

Rick, B., McGrath, D., Armstrong, W., & McCoy, S. W. (2022). Dam type and lake location characterize ice-marginal lake area change in Alaska and NW Canada between 1984 and 2019. *The Cryosphere*, *16*, 297–314. https://doi.org/10.5194/tc-16-297-2022

Sentinel Hub. (2024). Sentinel Hub. In *Sinergise Solutions d.o.o., a Planet Labs company* (No. v1.0.0). https://www.sentinel-hub.com

Van den Bossche, J., & others. (2024). Geopandas/geopandas: v1.0.1. *Zenodo*. https://doi.org/10.5281/zenodo.12625316

Wieczorek, I., Strzelecki, M. C., Stachnik, Ł., Yde, J. C., & Małecki, J. (2023). Post-Little Ice Age glacial lake evolution in Svalbard: Inventory of lake changes and lake types. *Journal of Glaciology*, *69*(277), 1449–1465. https://doi.org/10.1017/jog.2023.34

Wilkinson, M. D., Dumontier, M., Aalbersberg, Ij. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., Silva Santos, L. B. da, Bourne, P. E., Bouwman, J., Brookes, A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers, R., … Mons, B. (2016). The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, *3*, 2052–4463. https://doi.org/10.1038/sdata.2016.18

DRAFT